

Multi-Agent AI Orchestration Platform — Assignment 2

Services, Contracts & Schema

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Git branches: feature/frontend → Harshita; feature/backend → Harendra; feature/agents-db → Mohit; feature/testing-infra → Anurag.

1. Capabilities

#	Capability
1	Register
2	Login / Authenticate
3	Manage Profile
4	Create Workspace
5	View Workspaces
6	Manage Workspace Members
7	Create New Task
8	View Task Details
9	Validate Task Input
10	Run Task
11	Plan & Decompose Task
12	Assign Work to Agents
13	Execute Specialized Agents
14	Share Agent Context
15	Review & Validate Results
16	Store Task & Results
17	View Task History & Logs
18	Manage Agents / Configure Prompts
19	View System Logs & Analytics
20	Deliver Final Output

2. Service Design & Ownership

The use-case diagram is used as the functional basis. For Assignment 2, those capabilities are grouped by data ownership. The frontend branch is treated as the client/presentation layer rather than a data-owning service; it consumes the backend contracts. This keeps the rule that no persistent data is shared between services.

Service	Member / Branch	Data it owns
Identity & Access Service	Harendra Godara feature/backend	Users, credentials, roles, authentication sessions/tokens, access records
Workspace & Task Service	Harendra Godara feature/backend	Workspaces, workspace members, tasks, task dependencies, workflow runs
AI Agent Service	Mohit Tailor feature/agents-db	Agent definitions, agent runs, messages, shared agent context, agent outputs
Execution & Artifact Service	Anurag Sharma (Leader) feature/testing-infra	Artifacts, reviews, execution logs, run history, test/infrastructure records

Frontend / UI — Harshita Rathore — feature/frontend: client layer for login, workspace UI, task submission, status display, results and artifact display. It does not own persistent business tables in this benchmark.

Cross-service calls

Caller → Callee	Operation	Purpose
Workspace & Task → Identity & Access	checkAccess	Authorize a user for a workspace/task action
Workspace & Task → AI Agent	executeTask	Start an agent workflow for a task
AI Agent → Execution & Artifact	saveOutput	Persist agent outputs/artifacts
Execution & Artifact → AI Agent	requestReview	Request/complete agent review
Frontend → Workspace & Task	runTask	Submit and track the central task operation
Frontend → Identity & Access	loginUser	Authenticate the user

3. Service Contracts

Identity & Access Service

Operation	Input	Output	Errors
registerUser	name, email, password	user profile, registration status	invalid details; email already registered
loginUser	email, password	authentication result, access token	invalid credentials; account unavailable
checkAccess	user, resource, action	allowed / denied	resource not found; insufficient permission
getProfile	authenticated user	profile information	user not found

Workspace & Task Service

Operation	Input	Output	Errors
createWorkspace	workspace name, owner	workspace creation result	invalid name; owner not authorized
createTask	workspace, task description	task creation result	workspace unavailable; invalid task
runTask	task, execution request	run reference, status	access denied; invalid task; task already running
getTaskStatus	task / run reference	status, progress	task or run not found
cancelTask	task / run reference	cancellation result	already completed; cancellation unavailable

AI Agent Service

Operation	Input	Output	Errors
selectAgents	task description, available context	selected agent set	no suitable agent; context unavailable
executeAgent	run request, agent, context	agent output, execution status	agent unavailable; model failure; timeout
shareContext	run, context data	context update status	invalid context; run not found
requestReview	run, outputs	review request/result	review agent unavailable; invalid output

Execution & Artifact Service

Operation	Input	Output	Errors
saveOutput	run, output, artifact metadata	artifact result	invalid artifact; persistence failure
getArtifacts	run reference	artifact list	run not found
saveReview	run, review result	review status	invalid review; persistence failure
getRunLogs	run reference	ordered execution logs	run not found

4. Central Operation — runTask

Purpose: Execute a user's complex task through the Multi-Agent AI workflow and return a reviewed final result.

Aspect	Specification
Inputs	Authenticated user context, workspace context, task description and execution options.
Outputs	Run status, progress, reviewed result and references to stored artifacts/logs.
Error cases	Access denied; invalid task; task already running; no suitable agent; agent/model failure; timeout; review failure; persistence failure.
Internal details hidden	Task decomposition strategy, agent-selection rules, prompts, model/provider details, retries, internal messages, database schema, logs.
Internal flow	Validate access → validate task → create workflow run → plan/decompose → select agents → execute agents → share context → review → save review

5. Service Validation

Service	Reachable	Self-contained	Contract	Independent	Loosely coupled	Fix if needed
Identity & Access	Yes	Yes	Yes	Yes	Yes	—
Workspace & Task	Yes	Yes	Yes	Yes	Yes	—
AI Agent	Yes	Yes	Yes	Yes	Yes	—
Execution & Artifact	Yes	Yes	Yes	Yes	Yes	—

6. Database Schema

The ER design extends the same service boundaries. Each persistent table has one owner. Cross-service identifiers are represented as external references where a relationship crosses a boundary; the table itself remains owned by exactly one service.

Service	Owned tables
Identity & Access	users, auth_sessions
Workspace & Task	workspaces, workspace_members, tasks, task_dependencies, workflow_runs
AI Agent	agents, agent_runs, messages, agent_context
Execution & Artifact	artifacts, reviews, execution_logs